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4.3.3 Case 2: General case

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row space of A. So this is a vector in the row space of A. And V_R , the column space of V_R , is equal to the null space of A. So this is a vector in the null space of A. So we can call this here $\bar{x}$ and we can call this here x_n . Now if we take that and we plug that in here, what we get is $U_L \Sigma_{TL}$ times V_L Hermitian transpose times $V_L z_T$ plus $V_R z_B$ is equal to $U_L U_L$ Hermitian b . And if you now distribute

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Homework 4.3.3.1.

1/1 point (graded)
Reason that

$$\hat{x} = V_L \Sigma_{TL}^{-1} U_L^H b$$

is the solution to the LLS problem with minimal length (2-norm). In other words, if $x^\star$ satisfies

$$\|b - Ax^\star\|_2 = \min_x \|b - Ax\|_2$$

then $\|\hat{x}\|_2 \leq \|x^\star\|_2$.

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